

DUBAI MUNICIPALITY · WORLD'S FIRST AI-POWERED
PARK DESIGN CHALLENGE

ظلال الصفا

ZILAL AL SAFA

The first Dubai park *designed, simulated and operated* by an AI comfort brain — and how that intelligence shaped every stage of the design.

ZILAL AL SAFA

How artificial intelligence shaped every stage of the design — with human-led decisions

Dubai Municipality
AI Park Design Challenge
Plot 3570616 · 15,000 m²
Concept + Preliminary Design

01 Purpose & principle

The Challenge asks entrants to demonstrate how AI can be **meaningfully integrated throughout both the Concept and Preliminary design stages** — to support analysis, ideation, design development and decision-making — while maintaining human creativity and professional judgement. This report documents exactly that for ZILAL AL SAFA.

Our governing principle — “AI proposes, humans dispose.” At every stage, AI is used to **widen the option space** and to **quantify trade-offs** faster and more rigorously than manual methods allow. Every final spatial, cultural and aesthetic decision is made by the human design team. AI is a co-designer and analyst, never the author.

02 The pipeline at a glance

AI is applied through a **six-stage**, reproducible pipeline. Uniquely, it does not stop at the drawing: a sixth stage carries AI into **live operation** of the built park. Each stage has defined inputs, tools, AI outputs, the alternatives evaluated, and the human decision that closed it.

STAGE	QUESTION IT ANSWERS	PRINCIPAL AI ROLE
01 · Ingest	What is really happening on this site?	Data fusion, computer vision, NLP sentiment
02 · Generate	What layouts are worth considering?	Generative & parametric option search
03 · Simulate	How comfortable & usable is each option?	Environmental & pedestrian simulation
04 · Forecast	How will people actually use it?	Predictive experience & capacity models
05 · Visualise	How do we communicate it truthfully?	Diffusion-model imagery & data storytelling
06 · Operate	How does AI keep it comfortable after opening?	Live digital twin — the “Comfort Brain”

Every stage feeds the next; findings also loop back (e.g. simulation results re-seed a new round of generation). The loop is run until performance targets and design intent converge.

03 Data foundation

The analysis is grounded in the supplied competition material and open data, not assumption:

- **Supplied:** the DWG survey/base plan, the Neighborhood Parks Manual and the Scope of Work.
- **Site & context:** satellite & aerial imagery, GIS layers, cadastral geometry (90 × 166 m, plot 3570616), adjacent land uses (health centre, school, mosque, shopping, villas).
- **Behaviour:** Google “popular times” footfall curves (confirming the ~6 pm peak and midday collapse) and public review sentiment.
- **Climate:** Dubai sun-path, temperature, humidity and wind-rose normals for solar, shade and thermal-comfort modelling.

04 Stage-by-stage method

1 Ingest – site intelligence

Objective: build an evidence base of how the site performs and how people use it. **Tools/methods:** GIS & QGIS for spatial layers; computer-vision extraction of tree canopy, paths and hard/soft surfaces from aerial imagery; sun-path & wind-rose computation; NLP sentiment analysis of public reviews; parsing of the DWG survey.

AI outputs: a canopy & surface map, an hourly footfall curve, a heat/shade exposure baseline, and a ranked list of user-voiced issues (heat, shade, seating, facilities). **Human decision:** the team confirmed the core problem statement – “reclaim the midday heat-gap hours” – as the project’s north star.

2 Generate – layout option search

Objective: explore many spatial arrangements rather than one. **Tools/methods:** parametric modelling (Grasshopper-style rule systems) plus generative models to vary path geometry, canopy placement, zone adjacencies and program mix, subject to constraints (retain pitch/court & mature trees, four gateways, ~1 km loop, ~15% leasable).

AI outputs: dozens of candidate layouts, each scored on shade-reachability, walkability (route directness & loop length), program adjacency and retained-asset preservation. **Human decision:** the team selected and refined the “Cool Loop wrapping a cool social core” family, rejecting options that fragmented the canopy or pushed active uses into the quiet core.

Representative generation brief (paraphrased): “Generate park layouts on a 90×166 m plot that (a) provide a continuous shaded circuit ≥ ~1 km, (b) keep passive/social space in the thermally coolest zone, © retain the existing pitch, court and mature trees, (d) give each of the four edges a distinct entrance, and (e) reserve ~15% for leasable commercial use. Rank by shade-reachability and route directness.”

Stage-by-stage method (2)

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3 Simulate – comfort & flow

Objective: test how each shortlisted option actually performs. **Tools/methods:** hourly/seasonal shade ray-tracing and UTCI thermal-comfort modelling (Ladybug/Honeybee-type environmental tools) driven by Dubai climate data; pedestrian-flow modelling across the path network at the 6 pm peak.

AI outputs: per-hour shade maps, comfortable-area percentages (e.g. comfortable area at 3 pm July rising from ~9% to ~63%), and congestion/desire-line diagnostics. **Human decision:** canopies were concentrated on ray-traced hot-spots (“more comfort per dirham”) and the active edge was oriented toward the busy arrival, keeping the grove calm.

4 Forecast – experience & capacity

Objective: anticipate how the redesigned park will be used before it is built. **Tools/methods:** machine-learning models estimating dwell time, hourly capacity and inclusivity coverage by zone, calibrated to the Manual's targets (150–400 visitors / 10,000 m², 1–3 hour visits) and the observed footfall curve.

AI outputs: projected extension of comfortable usable hours (~+5.5 h/day in summer), balanced zone loading, and gap-detection for under-served user groups. **Human decision:** amenities (seating, WCs, water points, rest nodes ≤ 60 m apart) were distributed to remove the forecast gaps, and the events calendar was scheduled to the climate year.

5 Visualise – render & narrate

Objective: communicate the scheme truthfully and vividly. **Tools/methods:** diffusion-model image generation for day and night perspectives, art-directed by the team to match the plan; data-visualisation of the modelled results; assembly of the interactive microsite, board set and one-minute concept animation.

AI outputs: photorealistic concept perspectives (aerial, majlis, Cool Loop, all-abilities play, and a night view), the charts in this submission, and draft narrative text. **Human decision:** the team curated, corrected and captioned every image so visuals remain faithful to the measured design intent – renders illustrate the concept and are labelled as such.

Stage 6 — AI that operates the park

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6 Operate — the Comfort Brain DUBAI-FIRST

Most AI park entries stop at the drawing. ZILAL carries AI into **live operation**, so the finished park stays comfortable in real time — our single strongest differentiator on the 20% AI criterion.

Objective: keep the built park comfortable, safe and efficient every hour. **Tools/methods:** a live **digital twin** fed by on-site IoT sensors (temperature, UTCI, humidity, wind, air quality and footstep/footfall data, including the “Every Step Counts” kinetic tiles) applies the same UTCI logic used at design stage to the operating park.

AI outputs: real-time routing that guides visitors via the park app to the coolest available seat, lane or majlis; automated tuning of misting, irrigation and lighting to demand and weather; and an open KPI dashboard for Dubai Municipality. **Human decision:** operations staff set comfort/energy policies and thresholds; the AI executes and learns within them, improving each season.

The signature innovation stack

The Comfort Brain anchors four benchmarked signature innovations that make ZILAL unique on every side — each measured against real built parks (including Dubai Municipality’s own Quranic Park) so every claim holds up:

SIGNATURE INNOVATION	WHAT IT IS & WHY IT STANDS OUT
The Comfort Brain	Live AI digital twin that routes visitors to real-time comfort and auto-tunes the park — no Dubai park operates one.
Kinetic Mashrabiya canopy	Heritage screen reimaged as sun-tracking responsive shade over the Majlis Grove — iconic and unmistakably Emirati.
Net-positive solar canopy	PV spine shades the Cool Loop and generates more energy than the park uses — powering the tech and exporting surplus.
“Every Step Counts” kinetic floor	Footstep-energy tiles at the sports court & a Cool-Loop stretch; footfall doubles as a live sensor for the twin.

Costed honestly. The Comfort Brain and sensor network are low-cost, high-impact software + IoT. The one premium item — the footstep-energy floor — is deliberately limited to ~100–300 m² of hero zones and justified on **engagement, data and safety lighting rather than kWh**; the solar canopy carries the real energy load. All of it is sized to sit inside the AED 35M budget.

05 AI applied across all required areas

The Scope lists specific areas in which AI should be applied. ZILAL uses AI in each:

AREA OF APPLICATION	HOW AI WAS USED IN ZILAL
Site & context analysis	CV surface/canopy extraction, GIS fusion, DWG parsing
User & community analysis	Footfall curves, review-sentiment NLP, persona synthesis
Environmental analysis	Sun-path, shade ray-tracing, UTCI, wind modelling
Design ideation	Generative & parametric layout search
Design development & optimisation	Iterative scoring of shade, walkability, adjacency
Sustainability & resilience	Irrigation/water-balance & solar-yield estimation
User experience & decision-making	Dwell-time/capacity forecasting, gap detection
Visualisation & communication	Diffusion renders, data viz, animation, microsite
Live park operation	Real-time digital twin (“Comfort Brain”): routing, auto-tuning & a KPI dashboard

06 Toolchain summary

FUNCTION	REPRESENTATIVE TOOLS / TECHNIQUES	OUTPUT USED
Spatial data & GIS	QGIS, satellite/aerial imagery, DWG survey	Base plan, context & surface maps
Computer vision	Segmentation of canopy / paths / surfaces	Existing-conditions baseline
Behaviour & NLP	Popular-times data, review-sentiment analysis	Footfall curve, issue ranking, personas
Generative design	Parametric (Grasshopper-style) + generative models	Layout option set & scoring
Environmental sim	Ladybug/Honeybee-type shade & UTCI, wind	Shade & comfort maps, KPIs
Predictive modelling	ML dwell-time / capacity / coverage	Usage & inclusivity forecasts
Visualisation	Diffusion image models; charting; web	Renders, charts, microsite, animation

Tool names are representative of the methods applied at concept stage; equivalent industry-standard tools may be substituted during detailed design without changing the methodology.

07 Validation, limitations & assumptions

- **Modelled, not measured:** figures are concept-level estimates grounded in Manual targets, climate normals and public data. They will be validated with an on-site survey, metered baselines and detailed simulation in the next stage.
- **Human validation:** every AI output was reviewed by the design team; implausible or off-brief results were discarded, not published.
- **Reproducibility:** inputs, constraints and scoring criteria are documented so the pipeline can be re-run and audited.

08 Responsible-AI governance

- **Human accountability:** named designers own every final decision; AI outputs are advisory.
- **Bias & inclusivity:** forecasts were explicitly checked for under-served groups (People of Determination, seniors, children, women, teenagers) and gaps were designed out.
- **Data ethics:** only aggregate, non-personal, publicly available behavioural data was used.
- **Transparency:** AI-generated imagery is labelled as concept visualisation; no render is presented as a photograph of an existing condition.

Human creativity + AI rigour. The cultural and creative core of ZILAL — the Ghaf-tree identity, the majlis social heart, the “reclaim the lost hours” idea, the sense of place — is human. AI's contribution is to make that vision **measurably better and defensible:** to prove the shade works at 3 pm in July, that the loop is genuinely ~1 km and walkable, that the water and energy numbers add up, and that no user group is left out.

09 The differentiator

Most entrants will submit renders and a paragraph about AI. ZILAL submits a **documented, reproducible pipeline** where AI demonstrably shaped analysis, ideation, development, decision-making and communication – with human judgement retained throughout – and then goes one decisive step further: AI also **operates** the finished park through the live **Comfort Brain**. This is our strongest Dubai-first, and it is precisely what the “Integration & Effective Use of AI” criterion (20% of the score) rewards.

The Comfort Brain anchors a stack of four benchmarked signature innovations – the AI comfort twin, the sun-tracking **kinetic mashrabiya** canopy, a **net-positive solar** canopy, and the interactive **“Every Step Counts”** kinetic floor whose footfall doubles as a live sensor for the twin – making the park not only AI-designed but genuinely interactive and unique on every side.

10 From concept to preliminary design

The same pipeline carries into the Preliminary stage: generation is narrowed to detailed layout resolution; simulation is re-run at higher fidelity on the chosen scheme; forecasting informs amenity sizing and phasing; and visualisation produces the developed drawings, sections, boards and animation accompanying this submission.

STAGE	CONCEPT DESIGN USE	PRELIMINARY DESIGN USE
Ingest	Problem framing & baseline	Survey-calibrated data
Generate	Broad layout families	Resolved layout & details
Simulate	Comparative comfort screening	High-fidelity verification
Forecast	Usage direction	Amenity sizing & phasing
Visualise	Concept renders & microsite	Developed drawings, boards, animation
Operate	Comfort Brain concept & sensor plan	Commissioned digital twin & live dashboard

Summary: AI is not a garnish on ZILAL – it is the method by which a culturally-rooted, human-authored vision was made comfortable, inclusive, sustainable and buildable, and proven with numbers.

This report accompanies the ZILAL AL SAFA concept + preliminary design submission (interactive microsite, drawing set, board set, renders, data charts and one-minute animation).